

Nanyang Technological University

MH6805 Machine Learning in Finance

Group Project Guideline

General Information

1. The group project score is composed of **25%** group score + **75%** individual score.
2. Each group should be formed by **three (3)** students, while group size of **four (4)** is acceptable only when the class size is not a multiple of 3. The group size will be taken into account when assessing the project in breadth and depth, meaning that a four-student group is expected to accomplish more than a three-student group would. In the formula of calculating the project score, its 25% as group score will be evaluated based on project's merit regardless of group size while its 75% as individual score will be evaluated based on the individual efforts, where group size will matter.
3. When any group member communicates with others, including me, via email on behalf of the group, s/he must include all the group members in the c.c. list.
4. Every member in every group must contribute to the project. In every project deliverable, every group needs to indicate the contribution of each member, e.g. "All contributed equally" or "A, B, C contributed (part I, II, III) or (30%, 20%, 50%), respectively". Besides the contribution statement, individual efforts will be examined in the project presentation, especially in Q&A session. Therefore, the project presentation should be designed well such that every member will have a chance to speak.
5. Each group will be assigned with a discussion group which should prepare questions for the presentation. The discussion group will be given the poster/PPT of the presentation group prior to the presentation.
6. Details about the presentation schedule and the discussion group will be released in NTU Learn after I have reviewed all the project proposals.

Project Requirements

1. The purpose of this project is to provide students a hands-on experience to apply the machine learning methods to their own real dataset and problem. Students should choose at least **two** methods learnt from this course and compare the results. However, students are not restricted to the methods taught in the class. In other words, you can research some advanced methods, which I do look forward to. Moreover, the programming languages are not restricted to R/Python as well.
2. The data has to be a **real** dataset. Students need to demonstrate how the selected techniques are used to solve a real-life problem. If the data are sensitive or confidential, data can be re-labeled, recoded or even transformed so that the identity of the data will not be disclosed. We do not require you to share the dataset either as long as you can accurately describe it in the Final Report. However, again, the dataset must be **real**.

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3. Students **CANNOT** combine this project with that of another module, unless students clearly indicate their difference in terms of methodology, milestones, etc. and it is subject to approval. This matter is about (self-) plagiarism, which is not acceptable and leads to serious consequence. If you have any question about this, always inquire me.

Project Deliverables

1. For each of the following group project deliverables, each group only needs to submit once. Late submissions will be subject to grade deduction.
2. Students need to submit their group proposal by **23 September 2025**. The proposal should contain an introduction to your problem, description of your dataset, objectives, proposed methods to use, and any progress (preliminary results). The proposal **should NOT be more than 4 pages**.
3. The group project presentation is scheduled on **28 October 2025**. **All** presentation and discussion group members **must attend**. The presentation must be in English. We encourage students attend all presentations, but it is not compulsory.
4. Students need to submit a PPT for presentation **noon of one day before the presentation**.
5. Students need to submit their group final report, related codes, and dataset (if applicable) **two days after the presentation**. The final report should contain an introduction to your problem, description of your dataset, objectives, methods to use, findings, results comparisons, and conclusion/discussion. The final report including references **should NOT be more than 7 pages**. **Appendices are not counted toward the 7-page limit**.